Varun A. Paturkar

PhD Student, Computer Science and Engineering, CVIT, IIIT Hyderabad

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Research Interests

Safe and reliable perception and reasoning for autonomous driving in real-world conditions. My current work focuses on building and evaluating trustworthy vision-language models for driving.

Education

Ph.D., Computer Science and Engineering CGPA 8.75/10 Aug 2025 – Present
Center for Visual Information Technology (CVIT), IIIT Hyderabad *Hyderabad, India*
Advisor: Prof. C. V. Jawahar
Relevant coursework: Computer Vision, Mobile Robotics, Statistical Methods in Artificial Intelligence, Responsible and Safe AI Systems

B.E., Mechanical Engineering CGPA 7.37/10 Aug 2016 – Jul 2020
Sir M. Visvesvaraya Institute of Technology (Sir MVIT), Visvesvaraya Technological University *Bengaluru, India*

Publications

[1] **Varun A. Paturkar**, Shankar Gangisetty, C. V. Jawahar. *MotoVerse: A Benchmark for Predictive Perception of Motorized Two-Wheelers in Dense Unstructured Traffic*. IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), Datasets Track, 2027. (to appear)

[2] **Varun A. Paturkar**, Shankar Gangisetty, C. V. Jawahar. *MOTOR: A Multimodal Dataset for Two-Wheeler Rider Behavior Understanding*. IEEE International Conference on Robotics and Automation (**ICRA**), 2026.
[Project Page] [Paper] [arXiv] [Code] [Dataset]

[3] **Varun A. Paturkar**, R. Yadav, R. Kala. *Sequential Place Recognition using Semantically Enhanced Images*. *Multimedia Tools and Applications* (Springer), 2023.

Under review

[4] **Varun A. Paturkar***, Sreenivas Narasimha Murali, Sangeeth Sudhakar Nambiar, Shankar Gangisetty, C. V. Jawahar. *LadderDrive: Evaluating Granularity and Reliability of Vision Language Models in Driving Scenarios*. Under review, 2026. (* corresponding author)

Research Experience

PhD Researcher Aug 2025 – Present
CVIT, IIIT Hyderabad, advised by Prof. C. V. Jawahar *Hyderabad, India*

LadderDrive: Evaluating Granularity and Reliability of Vision Language Models in Driving Scenarios
In collaboration with Bosch Research India; under review [4]

- Designed an object-centric diagnostic benchmark that evaluates the same focal object within the perception, prediction and planning stages along two axes: granularity, through coarse-to-fine question chains within each stage, and reliability, through response consistency and invalid-input rejection.
- Built the benchmark from 80 scenes across BDD100K, nuScenes and IDD-X, with 2,006 focal objects and 31,102 human-verified annotations, and measured each axis with a Granularity Score, a Consistency Score (paraphrase, temporal and cyclic) and an Invalid-Input Rejection Score.
- Evaluated eight general-purpose and driving-specific VLMs and VLAs, showing that models with similar QA accuracy differ in granularity, consistency and rejection, and that from perception to planning they increasingly answer a fine-grained question correctly after failing the coarser one.

MotoVerse: A Benchmark for Predictive Perception of Motorized Two-Wheelers in Dense Unstructured Traffic
WACV 2027 [1]

- Created the first benchmark for anticipating a rider's next manoeuvre and forecasting the paths of surrounding vehicles, using only a single low-cost front camera that is realistic to deploy on two-wheelers (949 clips from 50+ hours of riding by 18 riders).
- Built a semi-automated annotation pipeline (YOLOv11 detection, ByteTrack tracking, human correction) that produced trajectories for 31,000+ road agents, and tagged each clip for camera shake, weak lane discipline and occlusion to show where models fail.
- Benchmarked 16 models, from video networks to VLAs; showed a rider's manoeuvre can be predicted with 72.6% accuracy from only its first half, leaving time for safety systems to react, and that off-the-shelf driving VLMs remain well behind fine-tuned models.

Research Intern Jan 2025 – Aug 2025
CVIT, IIIT Hyderabad, with Prof. C. V. Jawahar *Hyderabad, India*

MOTOR: A Multimodal Dataset for Two-Wheeler Rider Behavior Understanding
ICRA 2026 [2]

- Created the first large-scale multimodal dataset centred on the rider, recording what riders see (front, rear and helmet cameras), where they look (eye tracking), and how the vehicle moves (GPS and IMU): 1,629 sequences, 25+ hours from 16 riders in dense Indian traffic.
- Labelled 12 riding manoeuvres, 6 conventional and 6 unconventional, along with whether each manoeuvre was legal, so models can judge legality as well as behaviour.
- Showed that adding gaze and vehicle motion to video improves recognition, raising accuracy from 47.7% to 52.9% for behaviour and from 58.4% to 69.0% for legality; released the dataset and code.

Research Assistant (Part-time) Aug 2021 – Mar 2023
IIIT Allahabad, with Prof. Rahul Kala *Prayagraj, India*

Sequential Place Recognition using Semantically Enhanced Images
Multimedia Tools and Applications, 2023 [3]

- Co-developed a place recognition method using semantically enhanced features that outperformed several state-of-the-art methods on a self-collected dataset and the Oxford RobotCar dataset.
- Designed a convolutional autoencoder feature extractor trained with a custom weighted loss and transfer learning, and an attention-aware distance function that improved place matching.

Industry Experience

Machine Learning Engineer Jan 2021 – Dec 2025
Navajna Technologies *Hyderabad, India*

Driver Monitoring System Data Program (with Mercedes-Benz R&D India)

- Defined what data the in-cabin driver-monitoring models needed, and planned how it would be collected, including the sensor setup.
- Built efficient pre-processing and automated quality-check pipelines that kept 98% of the 800 TB collected usable for model training.
- Led a 15-member team across 5 countries to run the data collection.

Crowd Safety Analysis (with Prof. Dinesh Manocha)

- Analysed video of a music-festival crowd disaster: led a team estimating crowd density, speed and direction of movement, and wrote the report on the factors that likely caused it.

Avanta AI

- Architected the end-to-end avatar with voice- and face-cloning models; redesigned face-cloning inference for a 50% performance gain and deployed with Intel OpenVINO for 85% faster inference.

Highway Surveillance System

- Built a drone and survey-vehicle video system combining a VGG-16 classifier for NHAI KPIs with object detection to localise anomalies, and computed highway efficiency ratings from the results.

Talks & Presentations

Paper presentation: MOTOR, IEEE International Conference on Robotics and Automation (ICRA), Vienna 2026
Tutorial: Transformers, CVIT Summer School, IIIT Hyderabad 2026
Talk: "From Pixels to Actions", CIE-IIITH CV Labs Workshop (8th Ed.), CV for Industrial Manufacturing 2026

Awards, Service & Certifications

Travel Grant, IEEE Robotics and Automation Society (RAS), to attend ICRA 2026 2026
Reviewer: ICRA, IROS, WACV, ACCV
Certified Artificial Intelligence Professional, Defence Institute of Advanced Technology (DRDO), Pune 2022

References

Prof. C. V. Jawahar, IIIT Hyderabad [Profile]
Prof. Rahul Kala, IIIT Allahabad [Profile]
Prof. Dinesh Manocha, University of Maryland [Profile]
Alok Jain, Founder, Navajna Technologies [LinkedIn]